

Transformer-Based Sleep Stage Classification and Personalized Sleep Health Profiling

Using Fast SCFormer-U, Evidential Learning and Soft-Viterbi Sequence Decoding



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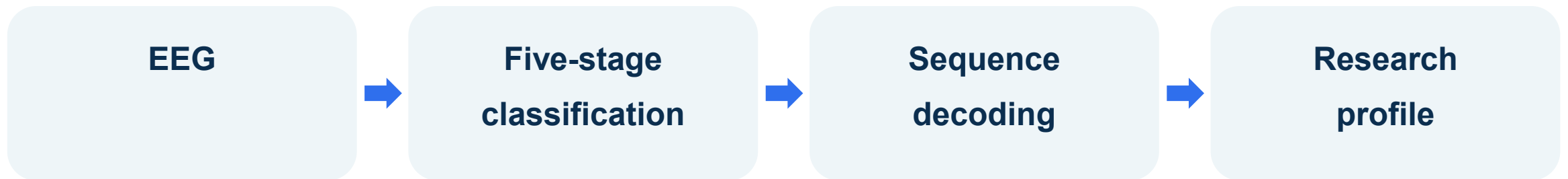
Enrollment No.: 23BIT044, 23BIT014

Department: Information and Communication Technology

Final evaluation: 7 October 2026

Problem Statement

- Classify four-channel EEG into Wake, N1, N2, N3 and REM at 30-second resolution.
- Automate epoch scoring, expose evidential uncertainty, and decode recording-level stage sequences.
- Convert scored sequences into interpretable research sleep profiles.



Scope: academic engineering research; no diagnostic or clinical-validation claim.

Project Objectives

01 Sleep-stage classification

Predict five stages using a compact CNN–self-attention–BiLSTM evidential model.

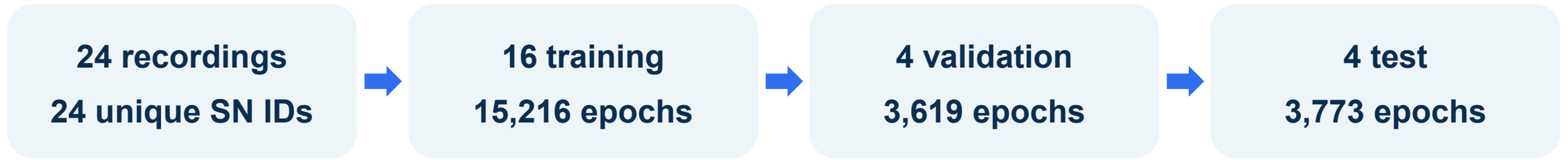
02 Sequence refinement

Decode stage probabilities with training-derived transitions, separately for each recording.

03 Subject-level profiling

Compute source-defined biomarkers, compare with ground truth and apply research risk rules.

Dataset and Experimental Split



Total evaluated usable epochs: 22,608

Overall counts: Wake 4,288 · N1 2,446 · N2 8,258 · N3 4,103 · REM 3,513

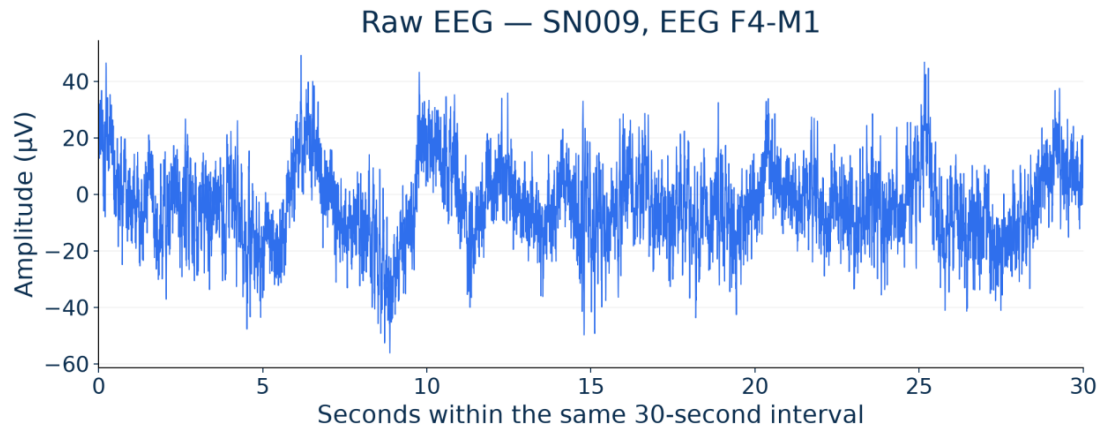
Selected EEG channels: F4–M1 · C4–M1 · O2–M1 · C3–M2

Test IDs: SN009, SN001, SN004, SN022

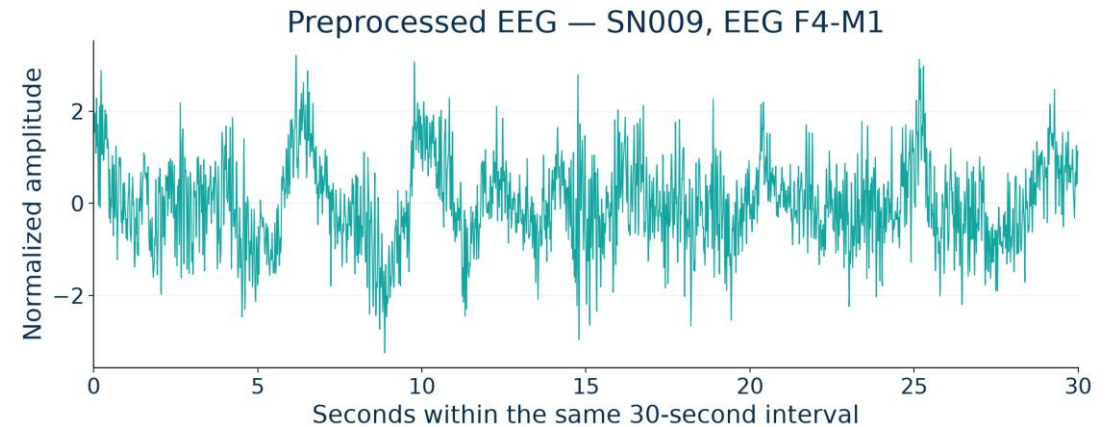
SN001 correction: 190 stored epochs → 854 EDF-derived epochs. Original data preserved.

No subject/recording ID appears in more than one split.

EEG Preprocessing



Recording interval 300–330 s. Original EDF at 256 Hz.



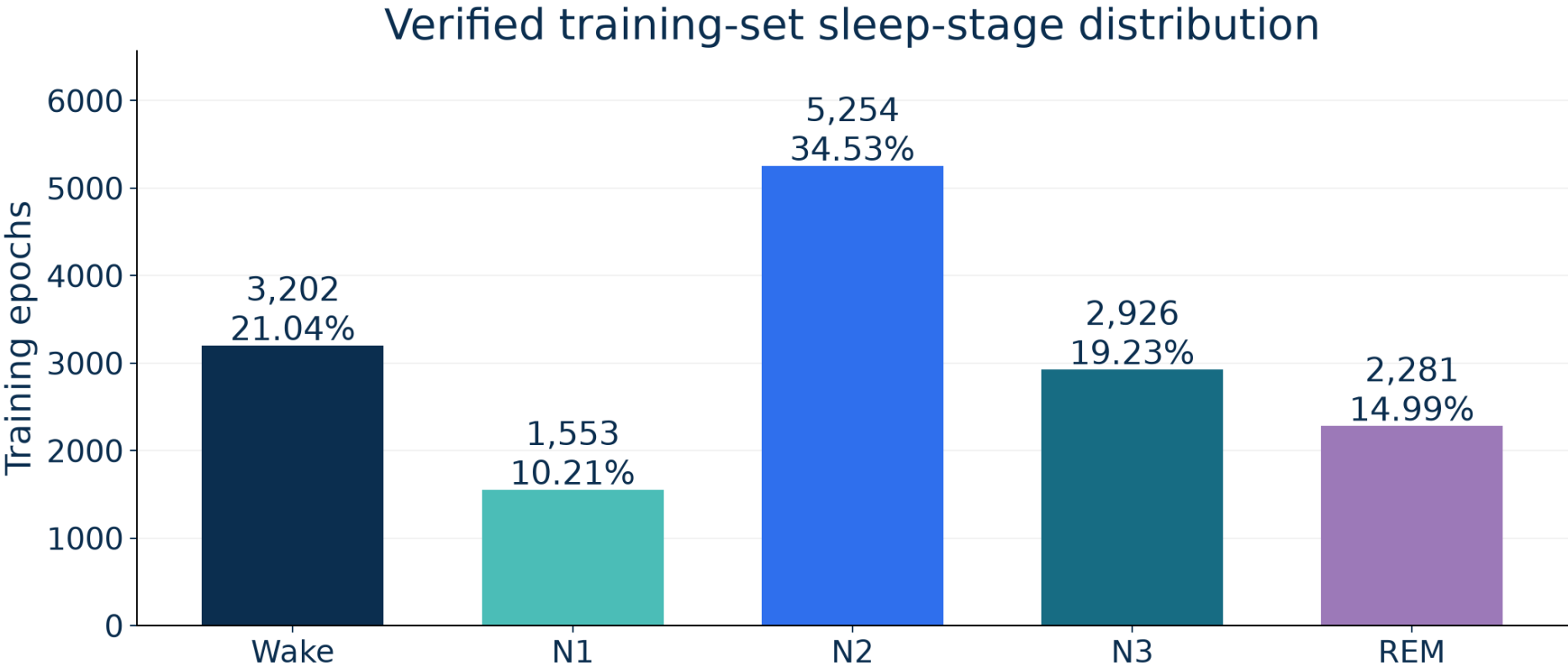
Recording interval 300–330 s. 50 Hz notch $\rightarrow$ 0.3–35 Hz bandpass $\rightarrow$ 100 Hz; per-channel epoch normalization.

Channel selection $\rightarrow$ 50 Hz notch $\rightarrow$ 0.3–35 Hz bandpass $\rightarrow$ resample 256 $\rightarrow$ 100 Hz

30-second epochs $\rightarrow$ channel-wise epoch normalization $\rightarrow$ input shape (4, 3000)

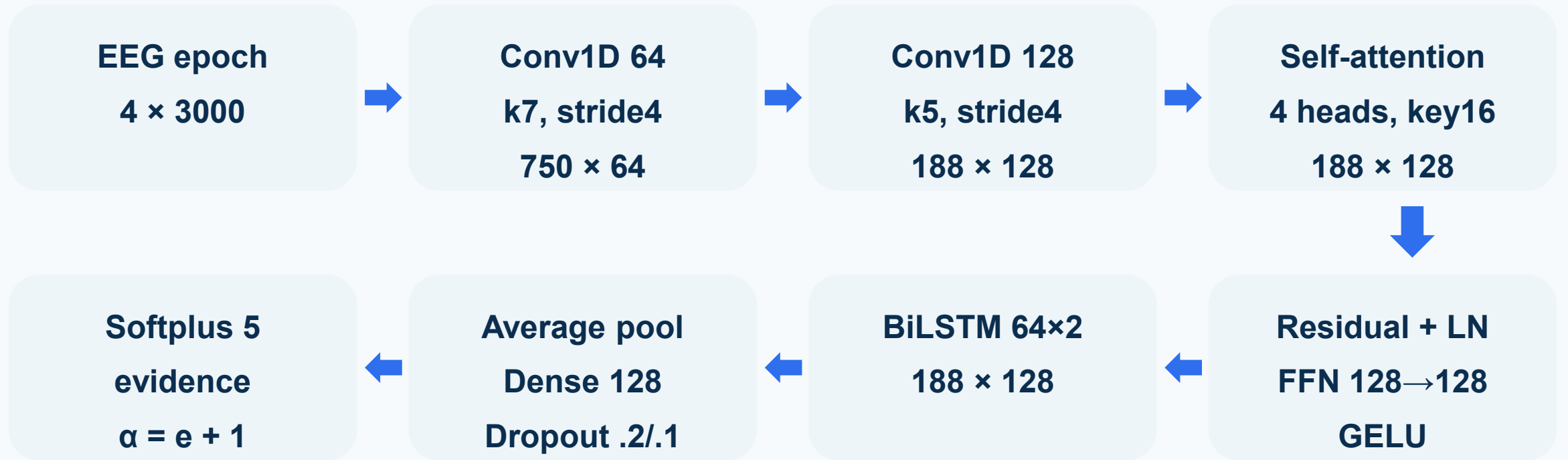
No amplitude-based rejection; `reject_by_annotation=False` in the source.

Sleep Stage Distribution



Class weighting: $N_{\text{training}} / (5 \times \text{training class count})$; computed from training labels only.

Fast SCFormer-U Architecture



Hybrid CNN-Transformer/self-attention-BiLSTM | 225,925 trainable parameters

Role of Each Model Component

CNN downsampling

Convolutional filters learn local patterns; strides reduce 3000 samples to 188 feature positions.

Self-attention + feed-forward block

Four attention heads combine feature positions within each individual 30-second epoch.

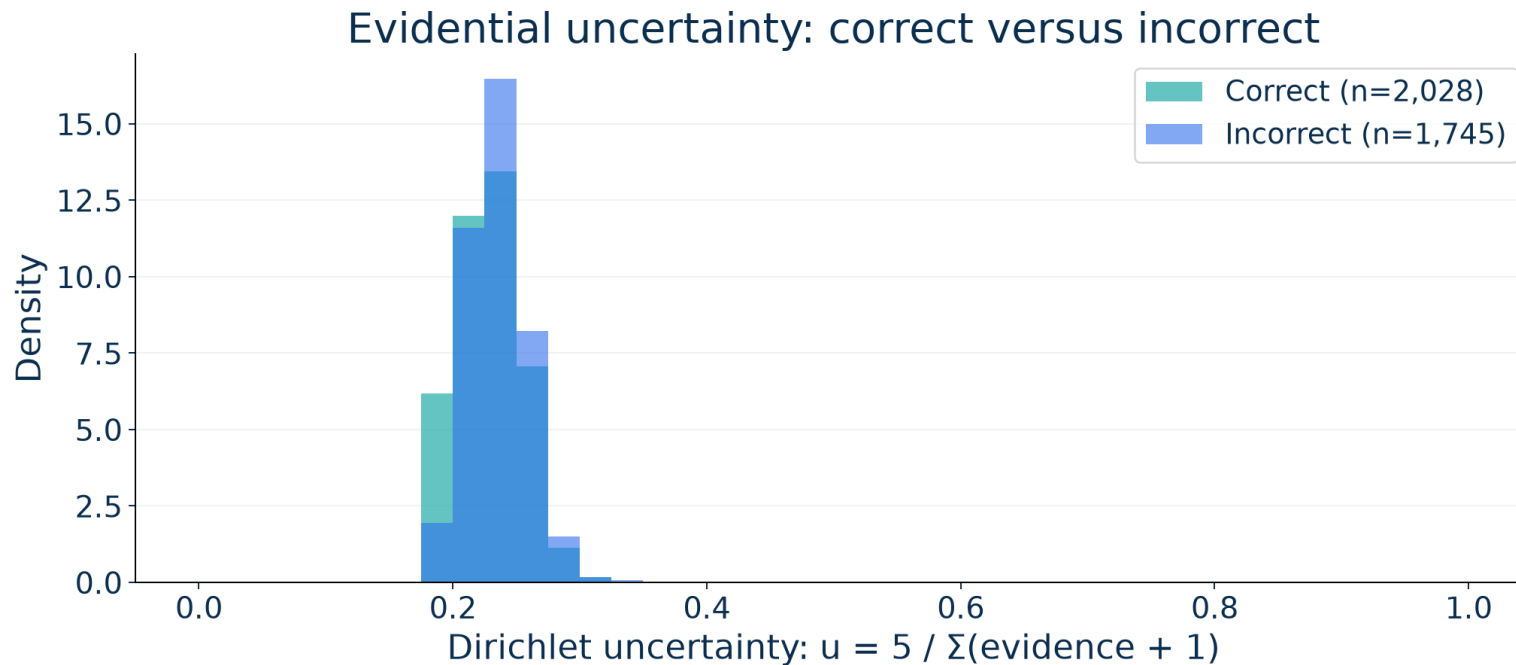
Bidirectional LSTM

64 units in each direction model the downsampled within-epoch sequence.

Evidential output

Five nonnegative evidence values define Dirichlet means and total-evidence uncertainty.

Uncertainty-Aware Classification



Mean uncertainty: correct 0.228; incorrect 0.235. Error AUROC 0.580.

Actual output

$$\alpha_k = \text{evidence}_k + 1$$

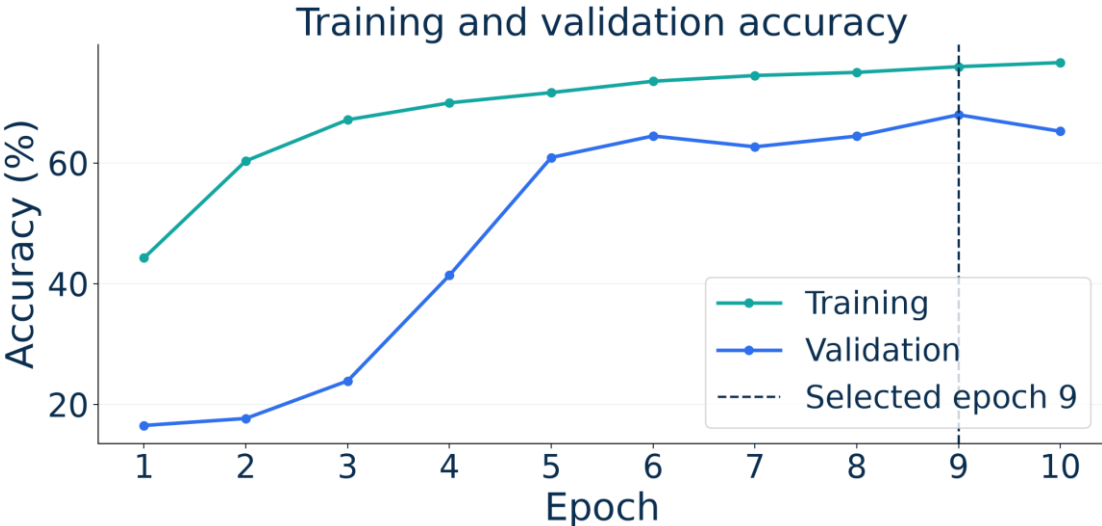
$$p_k = \alpha_k / \Sigma \alpha$$

$$u = 5 / \Sigma \alpha$$

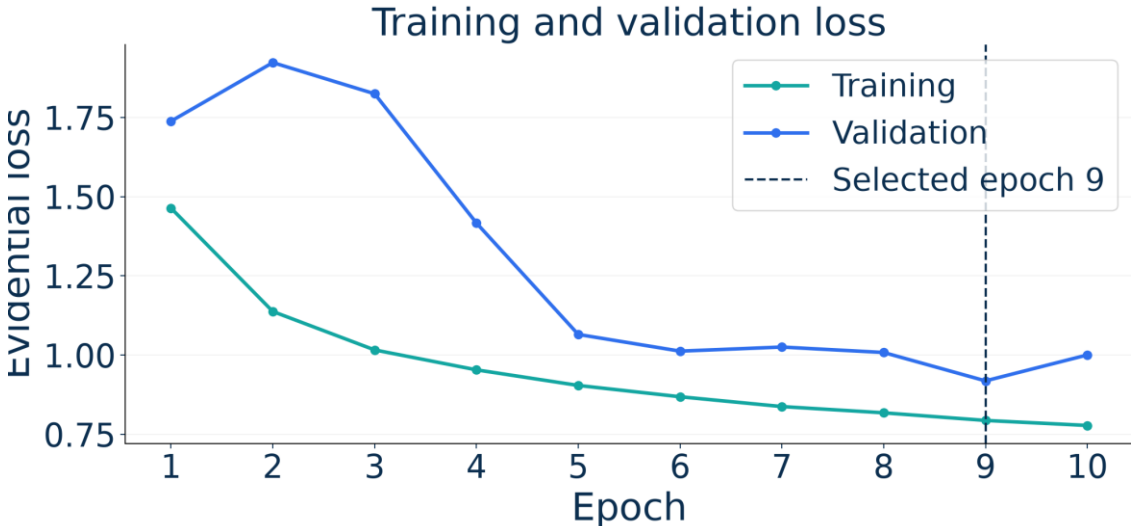
Error-detection AUROC
0.580

Higher uncertainty is useful only if it separates errors; this is evaluated, not assumed.

Model Training Progress



Saved training_log.csv; epoch 9 selected. Train loss uses class weights; validation loss is unweighted.



Saved training_log.csv; epoch 9 selected. Train loss uses class weights; validation loss is unweighted.

Setting	Verified value	Setting	Verified value
Optimizer	Adam	Learning rate	1×10^{-4}
Batch size	128	Epochs	10; selected 9
Early stopping	Patience 5; not triggered	Training hardware	Not preserved in run log

Sleep-Stage Classification Performance

Accuracy

53.75%

Balanced accuracy

57.83%

Macro F1

55.12%

Weighted F1

54.92%

Cohen's kappa

0.4060

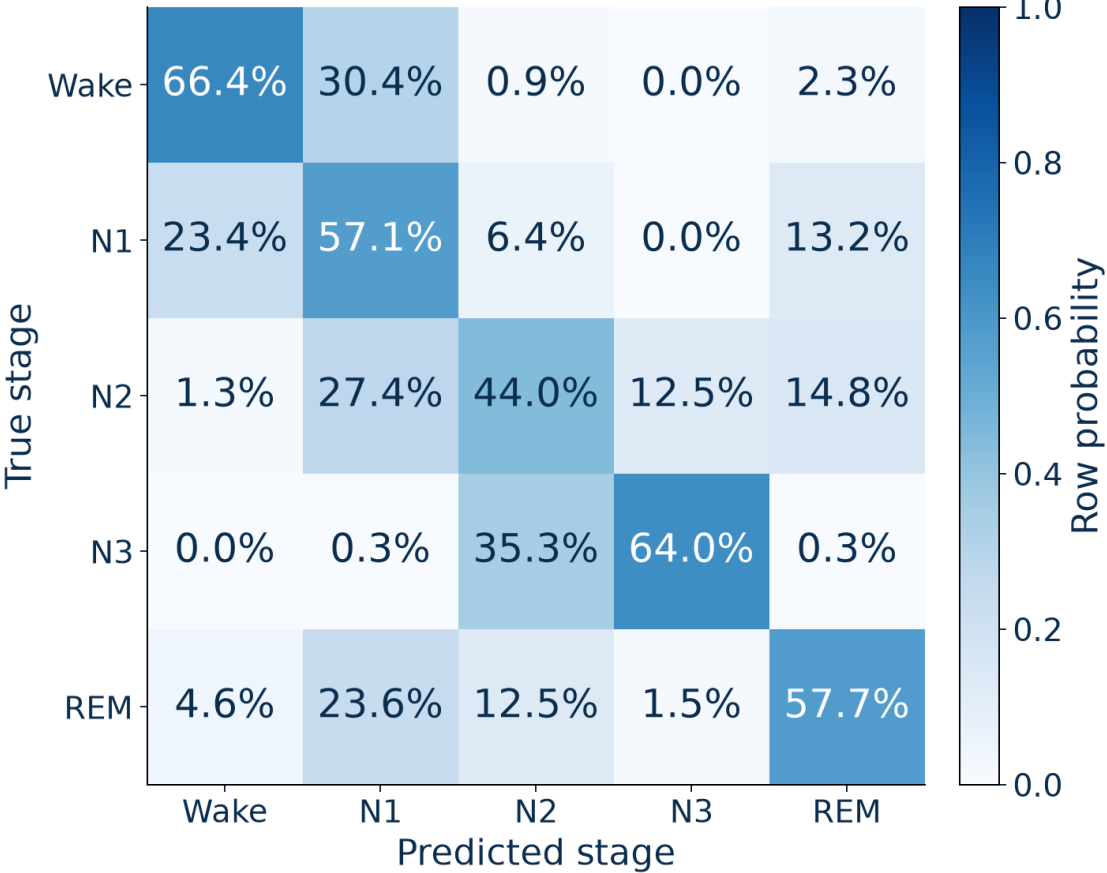
Raw evidential model

**3,773 epochs · four
held-out test IDs**

Final checkpoint, corrected test inputs, one run. Subject-disjoint fit; no external validation.

Sleep Stage Confusion Matrix

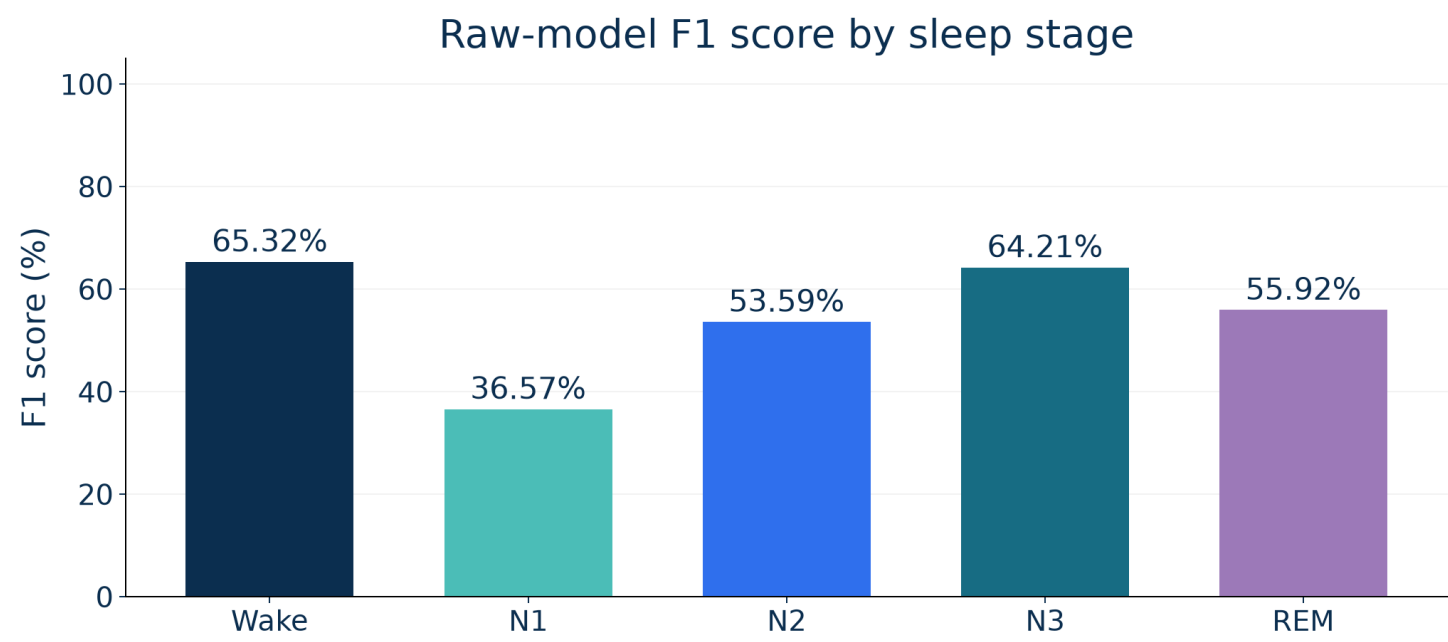
Raw evidential model: row-normalized confusion matrix



- Highest recall: Wake (66.36%).
- Lowest recall: N2 (44.00%).
- Largest off-diagonal share: N3 → N2 (35.33%).

Class order: Wake, N1, N2, N3, REM. HMC-PPT-20261007-01

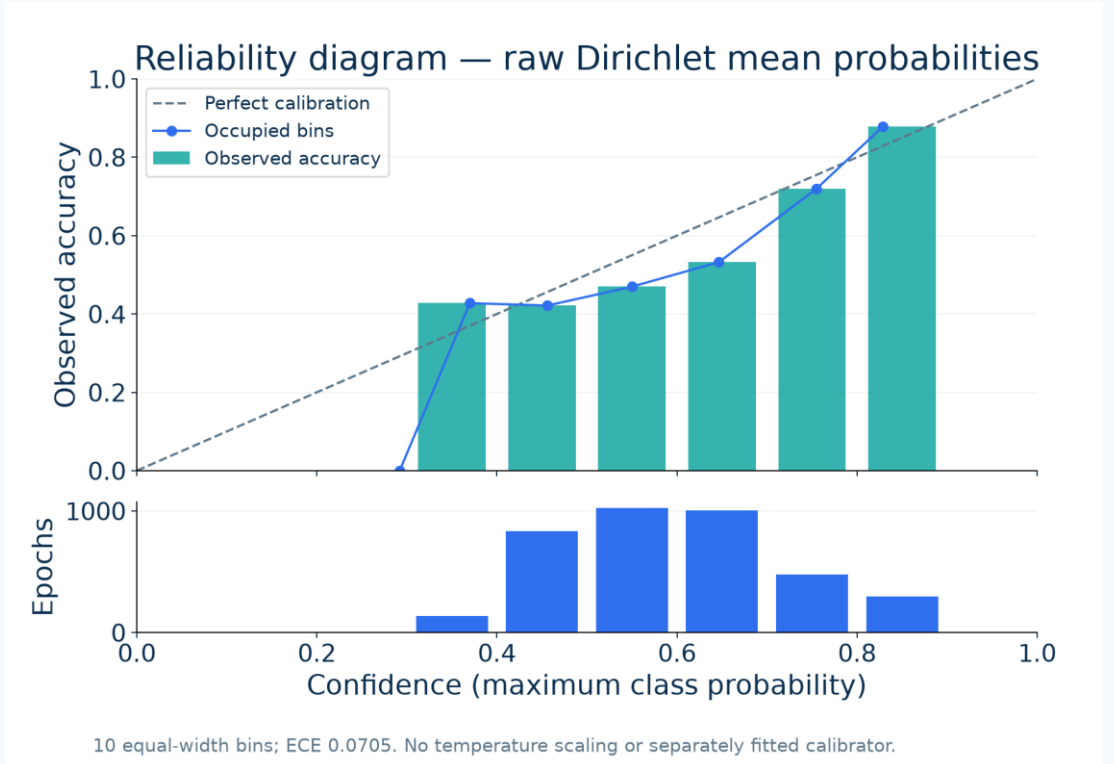
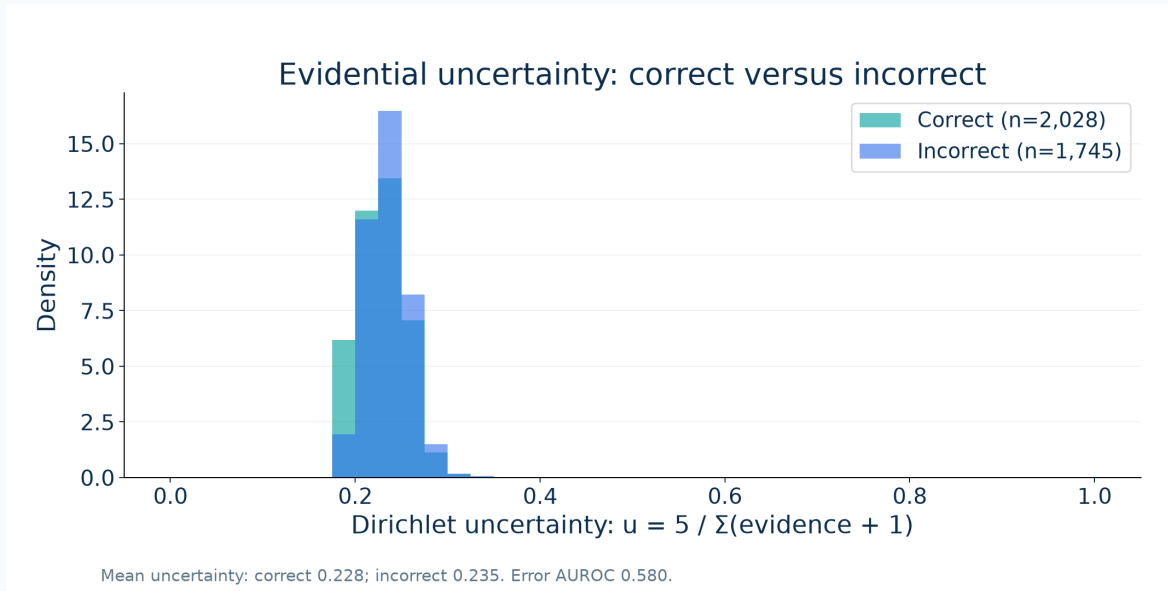
Class-Wise Classification Performance



Stage	Precision	Recall
Wake	0.643	0.664
N1	0.269	0.571
N2	0.685	0.440
N3	0.644	0.640
REM	0.543	0.577

Lowest class F1: N1; all values come from the raw-model condition.

Uncertainty and Calibration

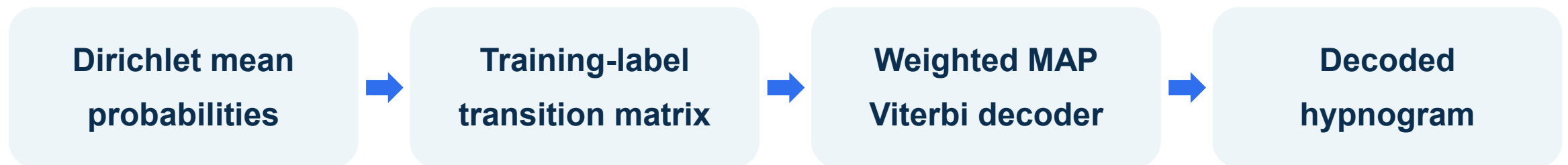


ECE (10 bins): 0.0705 | Brier (multiclass sum): 0.5879

NLL: 1.1012 | Error AUROC: 0.5798

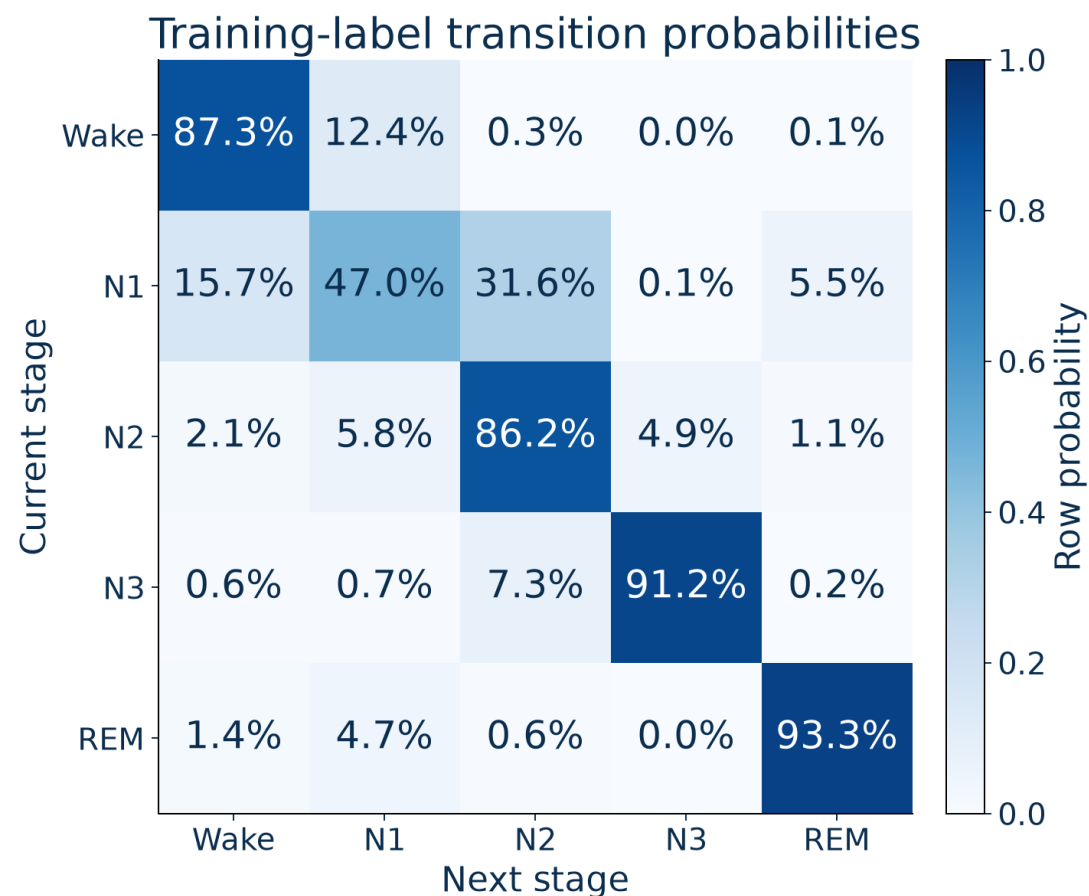
Soft-Viterbi Sleep Stage Refinement

- The model scores each 30-second epoch independently.
- The decoder balances those stage probabilities with transitions estimated from training labels.
- A log-space dynamic program selects one sequence for each recording; it never joins subjects.



Smoothing may improve aggregate scores while erasing brief stages. All transitions remain possible.

Sleep Stage Transition Modeling



Class order: Wake, N1, N2, N3, REM. Training labels only; pseudocount 1.

Training labels only

Pseudocount: 1 per cell

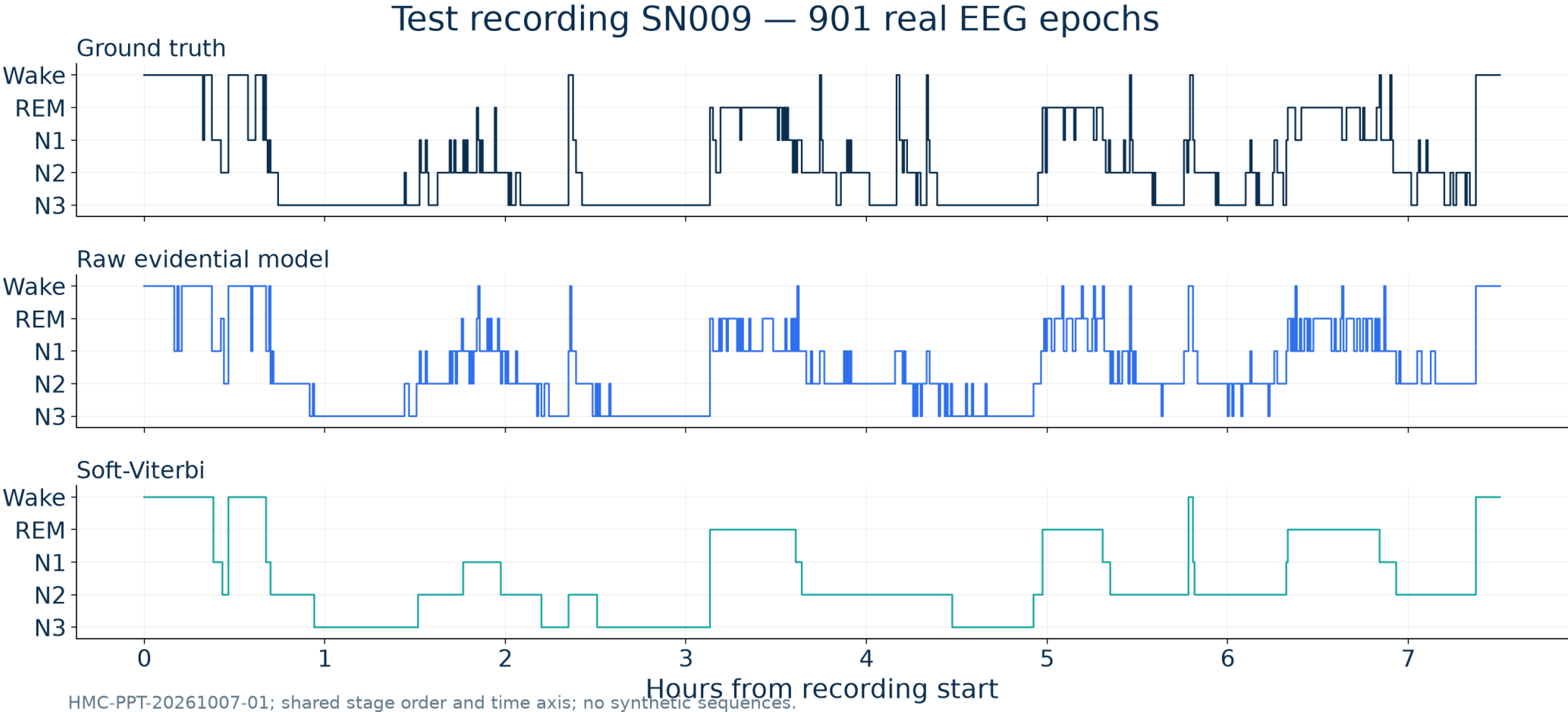
Rows sum to one

Initialization: stage prevalence

Separate recording boundaries

Emission log-weight: 0.9

Raw vs Smoothed Sequence — SN009



Impact of Sequence Smoothing

Metric	Raw model	Soft-Viterbi	Absolute change
Accuracy	53.75%	61.99%	+8.24 pp
Balanced accuracy	57.83%	64.26%	+6.43 pp
Macro F1	55.12%	61.76%	+6.64 pp
Weighted F1	54.92%	61.93%	+7.01 pp
Cohen's kappa	0.4060	0.4999	+0.0939

Improved: Accuracy, Balanced accuracy, Macro F1, Weighted F1, Cohen's kappa.

All listed aggregate metrics improved; per-subject outcomes may differ.

Personalized Sleep Health Indicators

Evaluated-recording SE (%)

$100 \times \text{non-Wake epochs} / \text{all evaluated epochs}$; a recording-duration proxy.

WASO (minutes)

Wake epochs between first and last sleep epoch $\times 0.5$; excludes trailing Wake.

N3 proportion (%)

$100 \times \text{N3 epochs} / \text{all evaluated epochs}$, including Wake in the denominator.

SFI (transitions / sleep hour)

N2/N3/REM $\rightarrow$ Wake/N1 transitions divided by hours scored as sleep; coarse stage-transition proxy.

No verified time-in-bed denominator; these indicators are research measures, not diagnoses.

Rule-Based Sleep Health Risk Stratification

Flags: SE <75% · WASO >60 min · N3 <10% · SFI >10 transitions/sleep hour

Priority	Research category	Exact assignment rule
1	High Risk	≥3 flags OR SE <65% OR SFI >15/h
2	Mild Risk	Otherwise, at least one flag
3	Healthy	Otherwise, no flags

Missing/nonfinite input → Unclassified in the new evaluator.

Thresholds are source-code rules; no clinical validation is established.

Test Subject Sleep Health Profiles

Recording	SE proxy	WASO (min)	N3 / all	SFI (/h)	Risk
SN009	90.01%	14.0	23.97%	1.18	Healthy
SN001	91.45%	31.5	24.71%	1.84	Healthy
SN004	88.48%	34.5	7.19%	2.27	Mild Risk
SN022	84.03%	74.0	9.08%	2.00	Mild Risk

Source: Soft-Viterbi labels only. Ground-truth profiles are stored separately.

Research category agreement with ground truth: 2/4 recordings.

SFI is a stage-transition proxy. Time in bed was not directly verified.

Complete End-to-End Framework



Second row is read right to left: probabilities → per-recording Viterbi → biomarkers → research rules.

Project Progress — Verified Deliverables

Completed and checked

Saved checkpoint loaded; disjoint split audited; corrupted SN001 test input reconstructed; inference rerun on all four test recordings.

Completed and checked

Confusion matrices, class reports, uncertainty/calibration, transition decoding, real EEG plots and all four hypnograms generated from one run.

Research limits remain

External validation, clinical threshold review and independent diagnostic validation have not been performed.

Current Research Limitations

- Limited cohort: 24 SN recording IDs; only four final test recordings (3,773 epochs).
- EEG-only inputs; no multimodal PSG or cross-dataset validation.
- Historical notebook test reuse limits claims of a newly blinded evaluation.
- Evidential uncertainty and reliability require stronger independent validation.
- Research biomarker definitions and thresholds are not clinically validated.
- Training hardware and final global training RNG state were not preserved.

Future Work

External validation

Evaluate on independent HMC recordings and another sleep dataset.

Multimodal PSG

Test EEG + EOG + EMG with a preregistered protocol.

Generalization

Study montage, device and population changes.

Calibration

Fit any calibrator on validation data, then test once.

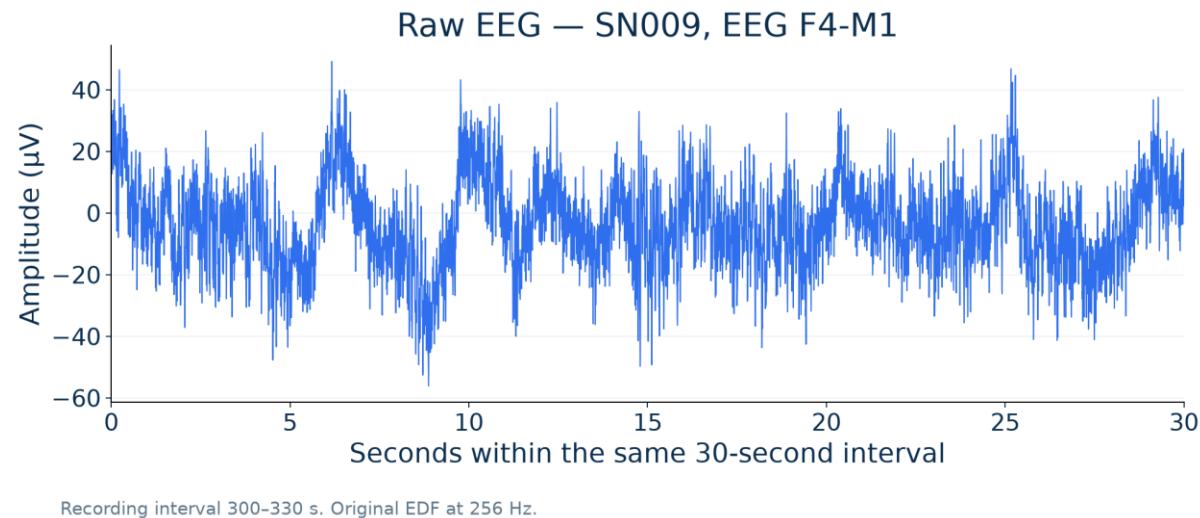
Clinical review

Review biomarker denominators and research thresholds with experts.

Prospective validation

Assess blinded recording-level outcomes before clinical claims.

Verified Example System Output



Test recording SN009

SE proxy: 90.01%

WASO: 14.0 min

N3 / all epochs: 23.97%

SFI: 1.18 /sleep h

Research category: Healthy

Outputs: hypnogram, epoch probabilities, uncertainty and ground-truth validation.

Conclusion

- A compact hybrid CNN–self-attention–BiLSTM pipeline supports five-stage EEG scoring and evidential uncertainty.
- On 3,773 epochs from four test recordings, raw accuracy was 53.75%; Soft-Viterbi accuracy was 61.99%.
- All final results use one checkpoint and the same verified test epochs; SN001's shortened input was corrected.
- Research sleep profiles are reproducible, but clinical validity and external generalization remain unestablished.

EEG → stage probabilities + uncertainty → decoded hypnogram → research sleep profile

Thank You

Questions & Discussion

Rikin Parekh · Om Ahuja
23BIT044 · 23BIT014
Project Guide: Dr. Santosh Sathpathy

Technical details and reproducible evidence

Checkpoint architecture · classification · uncertainty · sequence decoding

Training settings · source equations · additional real hypnograms · literature

Appendix — Model Summary

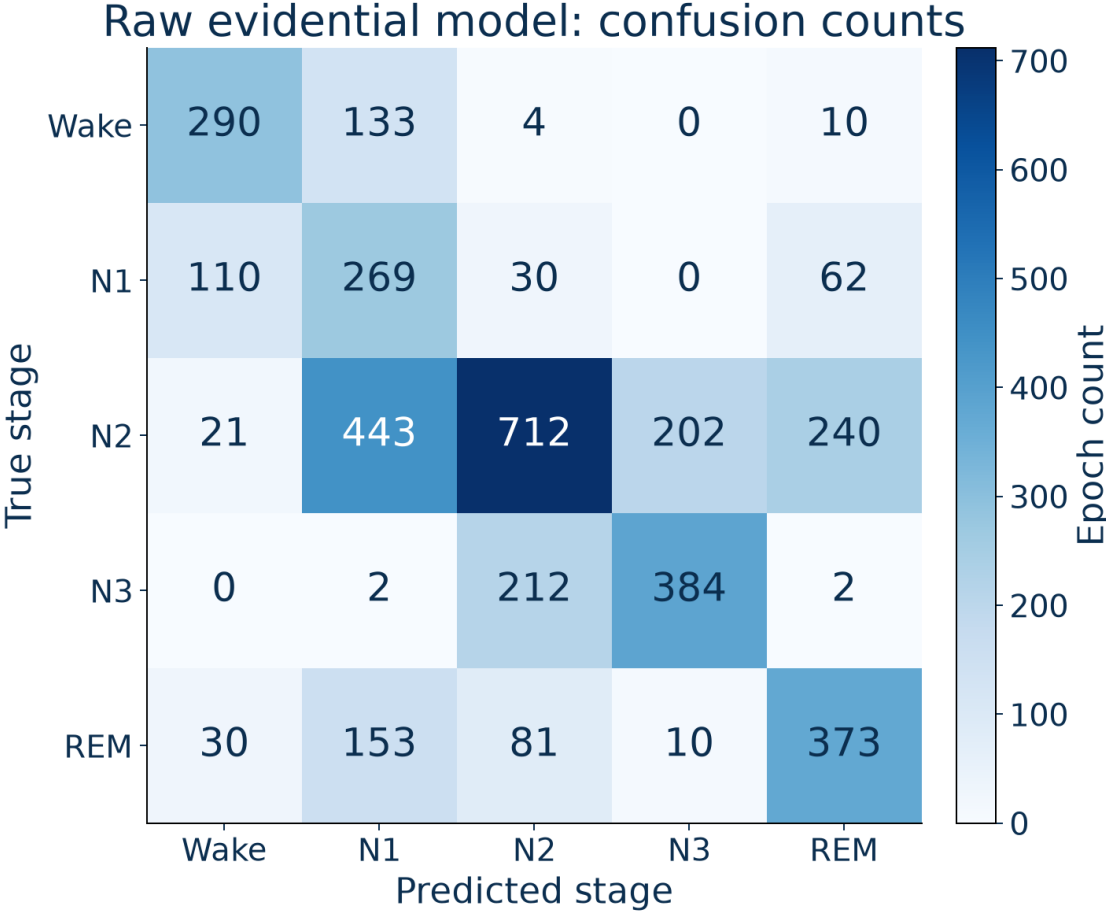
Component	Output / setting	Parameters
Input + permute	$4 \times 3000 \rightarrow 3000 \times 4$	0
Conv1D 64 + BN + ReLU	750×64	2,112
Conv1D 128 + BN + ReLU	188×128	41,600
Multi-head attention	188×128 ; 4 heads $\times$ key16	33,088
Residual/LN + FFN + residual/LN	188×128 ; GELU $128 \rightarrow 128$	33,536
Bidirectional LSTM	188×128 ; 64 per direction	98,816
Global average pool + dropout	128; dropout .2	0
Dense + dropout	128; ReLU; dropout .1	16,512
Evidential dense	5; softplus	645
Total 226,309; trainable 225,925; non-trainable 384. Full layer export in model_layers.csv.		

Appendix — Classification Report

Stage / average	Precision	Recall	F1	Support
Wake	0.643	0.664	0.653	437
N1	0.269	0.571	0.366	471
N2	0.685	0.440	0.536	1,618
N3	0.644	0.640	0.642	600
REM	0.543	0.577	0.559	647
macro avg	0.557	0.578	0.551	3,773
weighted avg	0.597	0.538	0.549	3,773

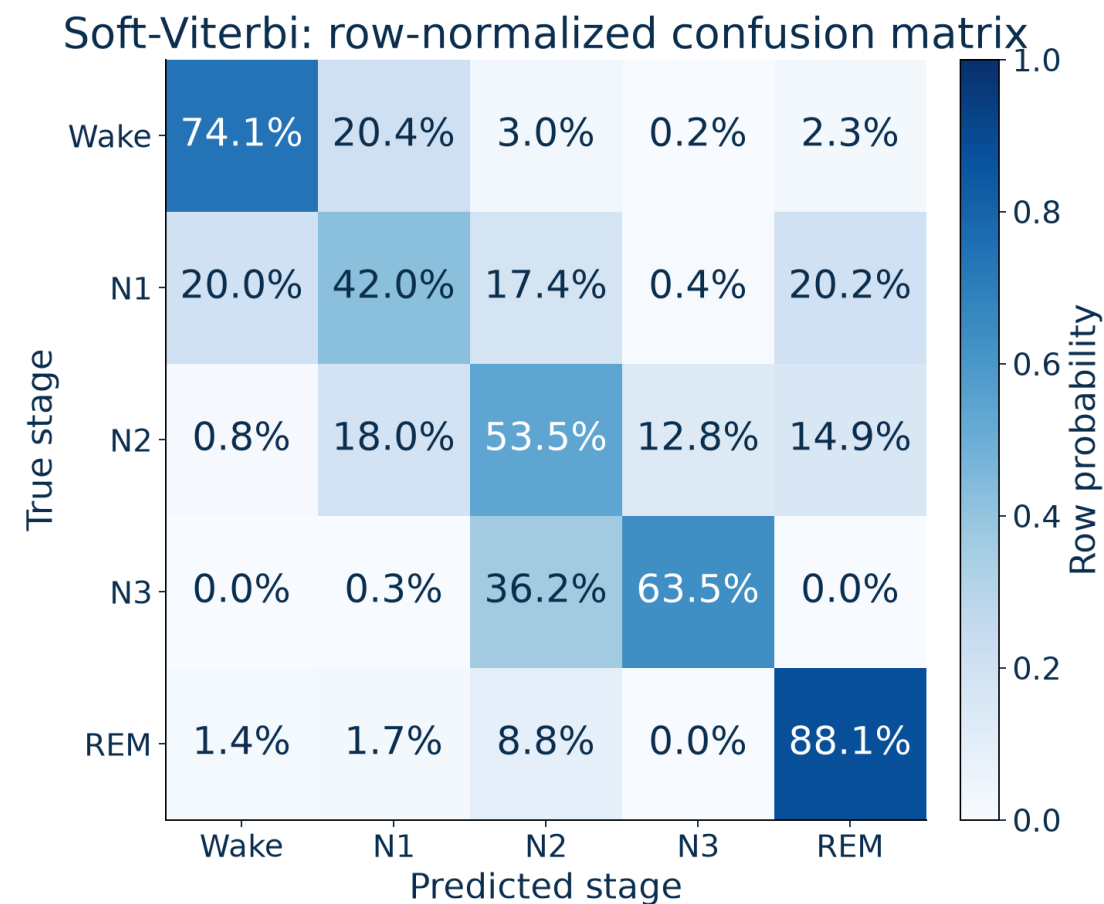
Raw model; support total 3,773. Macro averages weight stages equally; weighted averages use support.

Appendix — Raw Confusion Counts



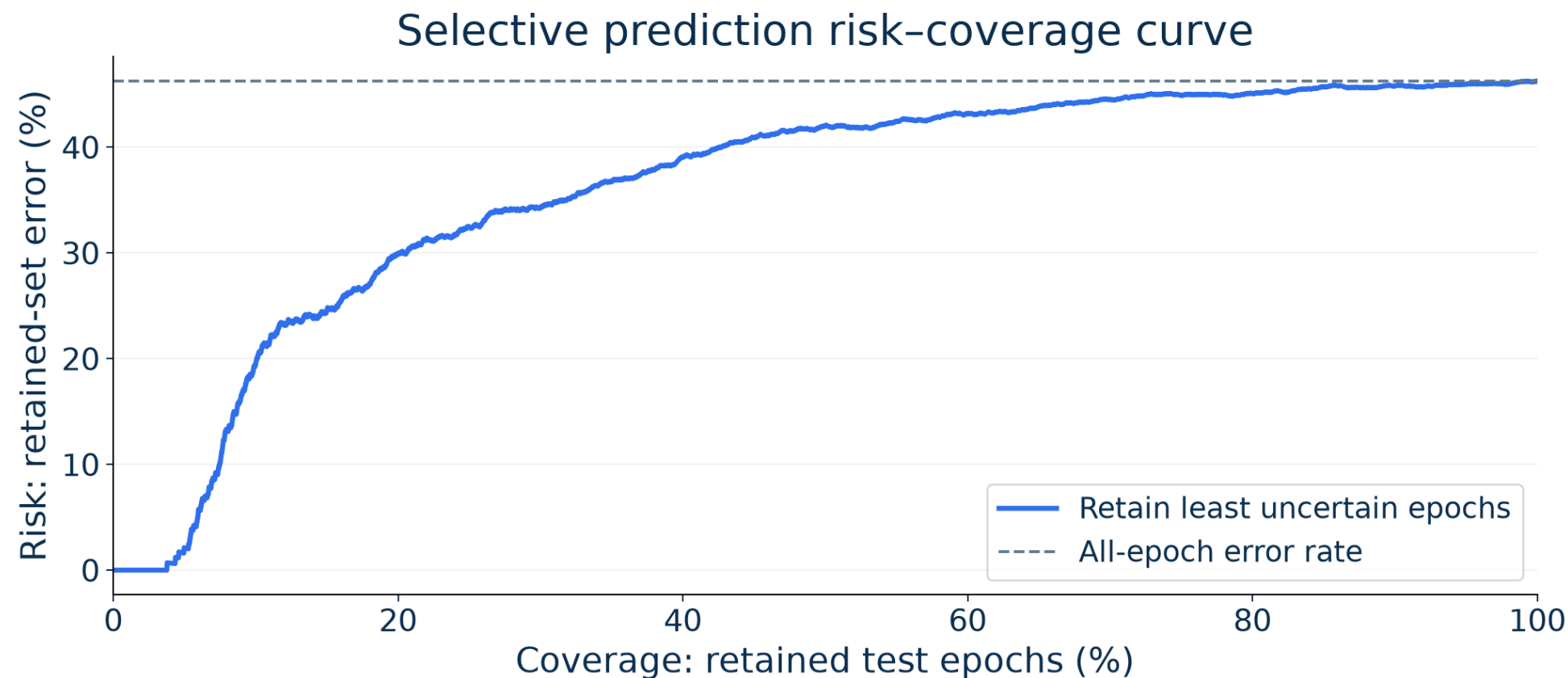
Class order: Wake, N1, N2, N3, REM. HMC-PPT-20261007-01

Appendix — Soft-Viterbi Confusion Matrix



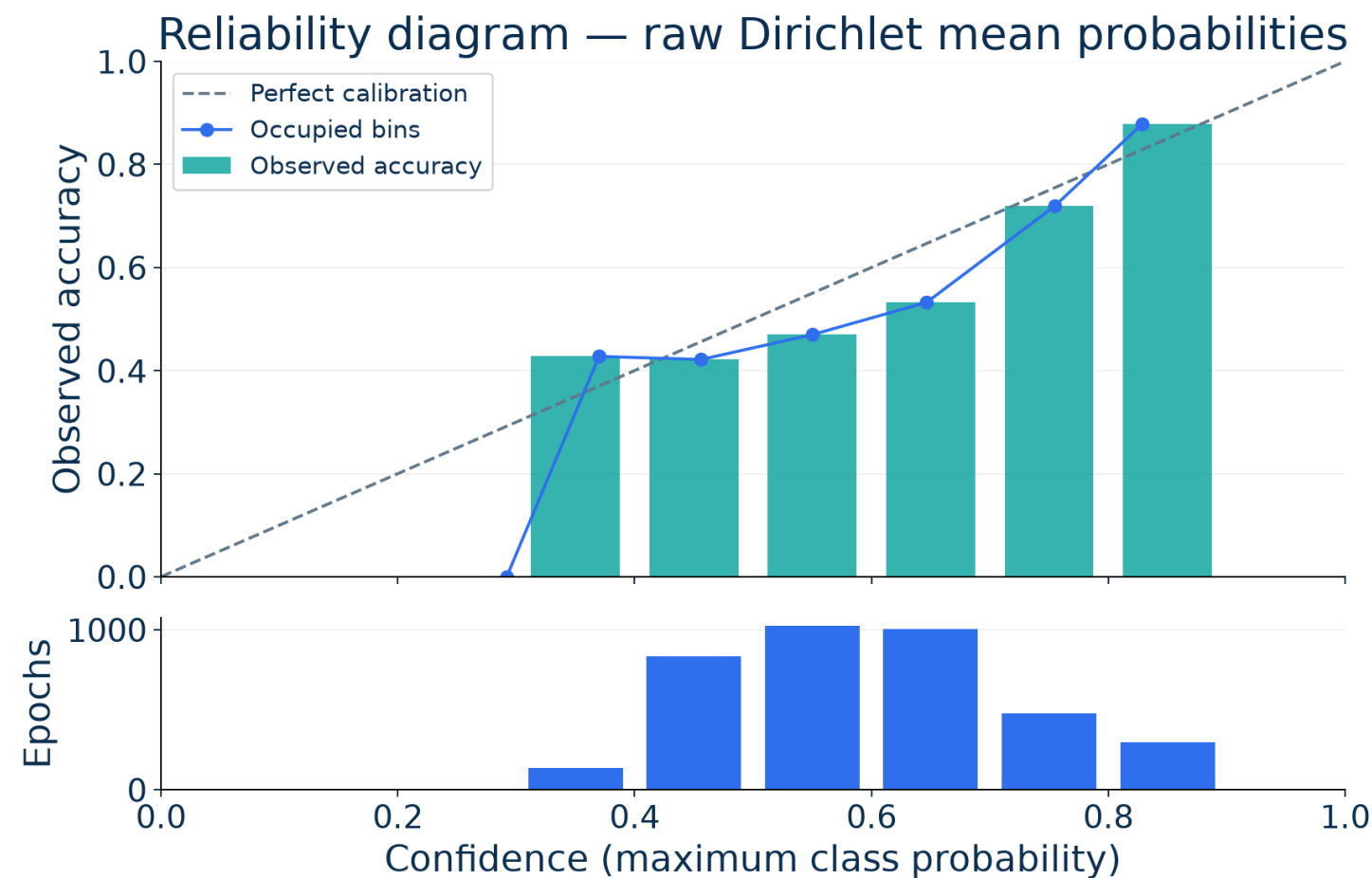
Class order: Wake, N1, N2, N3, REM. HMC-PPT-20261007-01

Appendix — Risk–Coverage Curve



Sort by $u = 5 / \Sigma \alpha$, ascending. AURC = discrete mean risk = 0.3628. No clinical risk interpretation.

Appendix — Reliability and Calibration



10 equal-width bins; ECE 0.0705. No temperature scaling or separately fitted calibrator.

Raw probabilities

10 equal-width bins

ECE: 0.0705

Brier: 0.5879

NLL: 1.1012

Bin counts shown below
the curve.

Appendix — Training Class Weights

Stage	Training epochs	Balanced weight
Wake	3,202	0.950406
N1	1,553	1.959562
N2	5,254	0.579216
N3	2,926	1.040055
REM	2,281	1.334152

$$wk = N_{\text{training}} / (5 \times N_k)$$

Balanced weights use training labels only; passed to `model.fit(class_weight=...)`.

Appendix — Final Hyperparameters

Setting	Verified configuration
Optimizer / learning rate	Adam / 1×10^{-4}
Batch / epoch limit / completed	128 / 10 / 10
Checkpoint selection	Maximum validation evidential accuracy; epoch 9
Early stopping / scheduler	Patience 5, restore best; not triggered / no scheduler
Dropout / evidence penalty	0.2 and 0.1 / coefficient 0.003
Label smoothing / class weighting	0.05 / balanced training-label weights
Seed provenance	Split seed 42; final training RNG state not recoverable
Hardware / versions	Training hardware not recorded; checkpoint Keras 3.13.2

Appendix — Implemented Biomarker Equations

Measure	Exact source equation
SE proxy (%)	$100 \times \text{nsleep} / N$
WASO (min)	$0.5 \times \text{Wake count between first and last sleep}$
N3 / all (%)	$100 \times nN3 / N$
SFI (/sleep h)	$\text{Count}(\{\text{N2}, \text{N3}, \text{REM}\} \rightarrow \{\text{Wake}, \text{N1}\}) / (\text{nsleep} / 120)$

N: evaluated 30-second epochs; nsleep: non-Wake epochs; nN3: N3 epochs.

WASO excludes trailing Wake; no-sleep WASO/SFI return 0 in source.

No verified time-in-bed denominator. SFI is a stage-transition proxy, not micro-arousal scoring.

Appendix — Weighted Viterbi Details

$A_{ij} = (1 + \text{training transitions } i \rightarrow j) / \text{row sum}$

$\pi_j = (1 + \text{training count}_j) / (5 + N_{\text{training}})$

$p_{tj} = (e_{tj} + 1) / \sum_k (e_{tk} + 1)$

$D_{0j} = \log(\pi_j + \epsilon) + 0.9 \log(p_{0j} + \epsilon)$

$D_{tj} = \max_i [D_{t-1,i} + \log(A_{ij} + \epsilon)] + 0.9 \log(p_{tj} + \epsilon)$

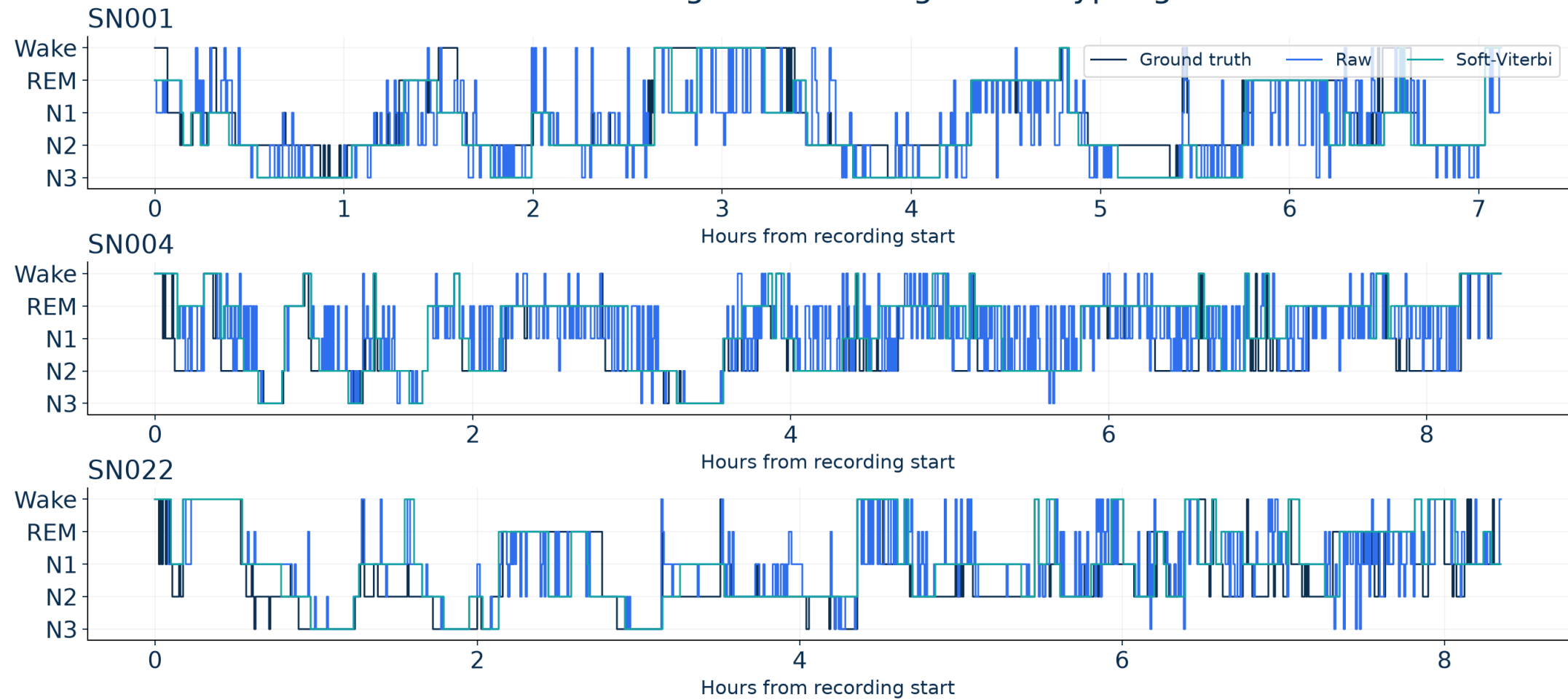
Store argmax predecessors; backtrack from $\arg\max_j D_{T-1,j}$.

$\epsilon = 10^{-12}$; training-only transitions; class order Wake/N1/N2/N3/REM.

Restart at each recording or missing-epoch gap. This is weighted MAP Viterbi, not differentiable soft-DP.

Appendix — Additional Test Hypnograms

Additional test recordings — overlaid genuine hypnograms



Overlay legend applies to all panels; stage order Wake, REM, N1, N2, N3. HMC-PPT-20261007-01

Appendix — Related Literature

Study	Dataset	Modality	Protocol	Accuracy / macro F1
SeqSleepNet [1] 2019	MASS; 200	EEG+EOG+EMG	20-fold subject CV	87.10% / 83.30%
DeepSleepNet [2] 2017	Sleep-EDF; 20	EEG Fpz–Cz	20-fold subject CV	82.00% / 76.90%
Current raw model 2026	HMC subset; 24	Four-channel EEG	Fixed 16/4/4 split	53.75% / 55.12%

Context only: different datasets, modalities and split protocols prevent direct ranking.

[1] DOI: 10.1109/TNSRE.2019.2896659

[2] DOI: 10.1109/TNSRE.2017.2721116

HMC dataset: DOI 10.13026/t79q-fr32. Full sources and protocols in references.txt.